

OPTIMAL DISTRIBUTION SYSTEM PLANNING

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Abstract - Models to represent substation feed area, feeder voltage drop, feeder load distribution, cost of losses in the feeders and transformers are formulated in terms of the variable system parameters. Based on these models, objective functions are defined which are employed in arriving at optimal substation size, feeder loading limits and conductor sizes. The technique suggested greatly reduces the computational time and effort compared to the methods presently in vogue. The optimal solutions, obtained for typical secondary and primary distribution systems using the proposed technique are presented. To bring out the efficiency of the proposed technique, the computational requirements for the proposed method along with the simulation techniques are discussed. The proposed method is highly promising, since it is very fast, simple and easy to program.

INTRODUCTION

Distribution systems, based on the voltage levels are classified as secondary and primary systems. The low voltage network consisting of secondary distribution feeders, feeding the LT consumers and the secondary distribution substations (SDS) which transform the voltage from a HT level to an LT level constitute the secondary distribution system. The primary distribution system consists of the primary distribution feeders, feeding the secondary distribution substations and the HT consumers, and the primary distribution substations (POS) that transform the voltage from a subtransmission or transmission level to a primary distribution level. The network that feeds the primary distribution substations, transferring energy from transmission to distribution systems is named as subtransmission system.

Until recently, distribution system received less attention in most of the utilities, with the result that fewer attempts were made to appreciate the need for concentrated efforts to formulate systematic planning approaches for distribution systems. In the meantime, in developing countries like India, due to the government policies for accelerated rural development and crash programmes for rural electrification, the system demands recorded a steep increase. But the distribution network growth was not in proportion to the load growth and it was mostly haphazard. This resulted in increased system losses, voltage problems and unreliable supply to the customers along with loss of equipment and production. The isolated planning and haphazard growth also caused highly uneconomical investments, posing serious problems to the overall economy. This situation in the distribution system management accentuated the need for an optimal design of distribution systems.

Some useful attempts, in this direction started with defining the several distribution system parameters and establishing the relationship between them [1,2]. Later on, several attempts have been made in suggesting algorithms for optimization of distribution system parameters through simulation study [3-8]. The main limitation of these models is the large computational time requirement. Also, difficulties are experienced in incorporating the operating constraints arising out of the choice of system parameters.

Further, attempts have been made using transshipment and branch and bound techniques for optimal network routing, thereby sizing and locating the substations [9-12]. These methods are applicable only to such areas, like urban, where the future locations of sinks and sources are known in advance through urban development master plans. In distribution systems, covering rural areas, dominated mainly by agricultural activities, the location of future loads and their development are dependent on several extraneous factors and it is often very difficult to identify their locations in advance. These models will not be adequate for such areas.

Thus, this paper aims at presenting a model to optimize the substation feed area, load carrying limit of the feeders and the conductor size for feeders. The present model is developed using the parametric relationships established in [13,14,15]. The present model is applicable for both secondary and primary distribution systems. The technique, discussed, will be useful in the planning of both urban and rural distribution systems and it has practically no limitations of problem size.

NETWORK CONSTRAINTS

Service quality is the main constraint of distribution network planning. The quality of service, ensuring declared voltage to the consumers, is a function of several variables such as voltage drops in the lines and transformers, tap settings in the transformers, voltage received at the source stations in the grid etc. All these variables, except the voltage drop in the feeders are dependent either on the standards of manufacturing or on the behaviour of the transmission system. The voltage drop in the distribution feeders is solely dependent on the choice of conductor size, level of loading, type of load distribution, power factor and circuit voltage. Increase in feeder voltage drop will lead to more system losses while the investment would be less. On the other hand, low voltage drop in the feeders will result in less system losses, but will lead to more investment. Thus, choice of an economic value for voltage drop is a compromise between the capital investment and the annual recurring expenditure due to energy losses.

The economic level of system reliability is a function of outage costs. Higher reliability levels force increased capital investment. Optimal reliability levels can again be determined striking a compromise between the increased capital investment and the outage costs. However, in the present discussion, this aspect

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has not been considered.

In the choice of network parameters, a number of alternate solutions are always possible to serve the system demands, fulfilling the voltage constraint; the extreme ones being that (i) a given area is served with less number of substations having large feed areas with lengthy feeders and (ii) the area is served with more number of substations having less feed area with short feeders. The former solution involves less investment and more system losses while the latter solution would cost more and have less loss of energy. The optimal solution is the one for which the total system cost including the capitalized cost of energy losses is minimum.

FORMULATION OF MATHEMATICAL MODELS

Mathematical models to represent feeder voltage regulation, feeder load distribution, substation feed area, substation and feeder costs, feeder loss cost and transformation loss cost are developed first to facilitate the choice of objective functions, needed. The following assumptions have been made in formulating the models: (i) The load density is uniform in the area implying that all the consumers are having the same maximum demand and power factor and are situated at equal intervals. (ii) The system is balanced under steady state operating conditions having no loss or no voltage drop in the neutral wire. (iii) The customer take-off points are equally spaced along the secondary feeder and the laterals are taken-off on both sides of the main feeder at equal intervals with equal number of customers. (iv) The primary laterals are taken-off at equal intervals on both sides of the primary main, feeding equal number of secondary distribution substations. (v) The substations are at the load centre so that all the feeders, running in different directions are equal in length and carry equal loads. (vi) All the substations have equal number of feeders. (vii) Both the mains and laterals of all the feeders use the same conductor size. (viii) All the feeders are radial.

Unit Voltage Regulation Constant (H)

It is known from [2] that the percentage voltage drop in a feeder with n sections can be obtained in terms of circuit voltage, line parameters, power factor angle and the moment of the loads as in (1)

$$V = \frac{\sum_{i=1}^n (kVA_i \cdot km_i) (r \cos \theta + x \sin \theta)}{10 (kv)^2} \quad (1)$$

Defining H as the moment for a voltage drop of 1% in the feeder as in (2)

$$H = \frac{\sum_{i=1}^n (kVA_i \cdot km_i)}{V} \quad (2)$$

and substituting in (1), we get,

$$H = \frac{10 (kv)^2}{(r \cos \theta + x \sin \theta)} \quad (3)$$

Thus, H is a function of circuit voltage, line parameters and power factor angle and is independent of percentage voltage regulation, feeder length and load in the feeder. It remains constant for a given conductor size, power factor and circuit voltage. It is named as unit voltage regulation constant and its unit is km-kVA.

Load Distribution Factor (LDF)

Since the voltage drop is also dependent on load, its distribution and feeder length, in addition to circuit parameters and power factor, it becomes necessary to define a factor, called Load Distribution Factor (LDF) as in (4)

$$LDF = \frac{(\text{Load in the feeder in kVA}) \times (\text{Feeder length in km})}{(\text{Moment of loads in km-kVA})} \quad (4)$$

Using the relationship in (2), (4) can be written as

$$LDF = \frac{PL}{Hv} \quad (5)$$

Equation (5) is true for both uniform and nonuniform load distribution. Equation (6) has been derived in Appendix A which gives the LDF for uniformly discretely loaded radial distribution feeders in terms of the number of lateral take-off points, n .

$$LDF = \frac{2n}{n+1} \quad (6)$$

Thus, LDF is independent of the variables given in (5) and is dependent only on the number of load tap-off points for uniformly loaded feeders. In Appendix A equations (7) and (8) have also been developed for two typical non-uniform cases, viz. (i) the spacing of laterals increasing from the far-end towards the substation in the arithmetic progression with equal load in each lateral and (ii) the spacing of laterals increasing from the substation towards the far-end in the arithmetic progression with equal load in each lateral. For case (i),

$$LDF = \frac{3n}{2n+1} \quad (7)$$

For case (ii),

$$LDF = \frac{n^2(n+1)}{2 \sum_{i=1}^n [i(n-i+1)]} \quad (8)$$

As can be seen from (7) and (8) that LDF is independent of the quantum of load, length or moment even with certain types of non-uniformity similar to the ones illustrated.

Further, it can be seen from (6), (7) and (8) that for uniformly loaded feeders, the value of LDF lies between 1 (for $n=1$, a feeder having a concentrated load at the far end) and 2 (for $n=\infty$, a feeder having continuously uniformly distributed loads along the feeder). It may also be observed that for a value of n equal to 10, LDF is 1.82, very close to its maximum value. It is, thus, possible to consider such feeders having the number of lateral take-off points more than 10 with equal loads as continuously uniformly distributed.

The first case of non-uniformity discussed in Appendix A (Fig. a) represents a system having most of the loads away from the substation. As n increases, more and more loads are pushed towards the far end. The extreme values of LDF in this case are 1 and 1.5. The second case (Fig. b of Appendix A) represents an opposite situation, having more loads closer to the substation. Increase in the value of n , moves the loads towards the substation. The value of LDF in this case lies between 1 and ∞ . However, the cases of these types corresponding to large value of n are not realizable in practice. The normal value of LDF for the above cases when n is 10 works out to 1.43 and 2.5 respectively.

Use of (5) for the practical systems, with a general non-uniform loading pattern, shows that the values of LDF for almost all the feeders in the area lie in a very close range. In many typical rural systems in

India, it has been found that the value of LDF lies between 1.8 and 2.2.

Thus, it is always possible to predetermine the value of LDF either from the existing system data or by selecting a desired pattern of distribution.

Substation Feed-area

The substation feed-area is dependent on the load density, number of feeders in the substation, voltage regulation, conductor size, load distribution, load diversity and power factor. The total load in the feeder is given by

$$P = \frac{a_s D}{n_f (PF)(DF)} \quad (9)$$

Substituting for P from (9) in (5), the substation area a_s is given by

$$a_s = \frac{Hv(LDF)(PF)(DF)n_f}{LD} \quad (10)$$

In a circular feed-area with radius R, the route of the main feeder is very rarely radial and its length is always more than the radius, R. This is mainly due to the geography of the area and the development of load. In a given system, the ratio, L/R remains more or less constant for all the feeders. This ratio is defined as zig-zag factor and is denoted by z. With the help of the above definition and (10), the radius of the circular feed-area, R is obtained as

$$R = \left[\frac{Hv(LDF)(PF)(DF)n_f}{\pi z D} \right]^{1/3} \quad (11)$$

In a similar way, for any given shape of the substation feed-area, equations to represent its dimensions can be developed by substituting for a_s and L in terms of its dimension into (10). For example, in a square feed-area of size $2a \times 2a$, a is obtained as

$$a = \left[\frac{Hv(DF)(LDF)(PF)n_f}{4 z D} \right]^{1/3} \quad (12)$$

Keeping all the variables in (11), except n_f and v, constant, (11) can be written as

$$R = k_1 (n_f v)^{1/3} \quad (13)$$

where

$$k_1 = \left[\frac{H(LDF)(PF)(DF)}{\pi z D} \right]^{1/3} \quad (14)$$

For a given conductor size, load density and load distribution in a system with constant load diversity and z, k_1 remains constant.

Substation and Feeder Cost

The substation cost is a function of the substation size and the number of feeder bays provided at the substation and can be represented as

$$c_s = e + g (\text{kVA}) + f n_f \quad (15)$$

f, the cost per feeder bay, is known independently, and e and g can be evaluated from the linearized cost versus kVA capacity characteristic of the substation excluding the bay cost. Typical values of e, g and f for secondary and primary distribution substations in a typical Indian system works out as

(a) Secondary distribution substation :

$$e = \text{Rs. } 6000, g = \text{Rs. } 105 \text{ per kVA and } f = 0$$

(b) Primary distribution substation :

$$e = \text{Rs. } 630,000, g = \text{Rs. } 60.38 \text{ per kVA and } f = \text{Rs. } 75,000.$$

The cost of a feeder with a given conductor size is only a function of its length as

$$c_l = c_f L \quad (16)$$

Transformation Loss Cost

The core loss and the full load copper loss of the transformer in kW bears a linear relationship individually with the transformer kVA capacity as in (17) and (18) respectively.

$$T_i = a' + b' (\text{kVA}) \quad (17)$$

$$T_c = c' + d' (\text{kVA}) \quad (18)$$

Typical values for a' , b' , c' and d' in the cases of 11/0.4 kV and 33/11 kV transformers are

(i) 11/0.4 kV transformers

$$a' = 0.06 \text{ kW}, b' = 0.0023 \text{ kW/kVA} \\ c' = 0.35 \text{ kW}, d' = 0.015 \text{ kW/kVA}$$

(ii) 33/11 kV transformers

$$a' = 0.725 \text{ kW}, b' = 0.001155 \text{ kW/kVA} \\ c' = 3.9 \text{ kW}, d' = 0.00605 \text{ kW/kVA}$$

The energy loss corresponding to the power loss in a transformer, given by (17) and (18) is obtained as

$$E_t = \left[T_i + T_c (UF)^2 (LLF) \right] 8760 \quad (19)$$

Assuming that n_t is the number of transformers in each of the n_s substations in the area A, the total transformation energy losses in the entire system is given by,

$$E_{st} = E_t n_t n_s \quad (20)$$

The cost of this energy loss is a recurring expenditure annually to the utilities. This annual expenditure can be capitalized by dividing the same by a capitalization factor, u, consisting of depreciation, interest, operation and maintenance charges. The capitalized cost of transformation losses are

$$C_t = E_{st} C_e / u \quad (21)$$

The variables kVA and n_s are functions of a_s as shown in (22) and (23) respectively.

$$\text{kVA} = \frac{D a_s}{(UF)(DF)(PF)} \quad (22)$$

$$n_s = A a_s^{-1} \quad (23)$$

Substituting for kVA, n_s , E_{st} , E_t , T_c and T_i from (22), (23), (20), (19), (18) and (17) respectively, C_t can be represented as

$$C_t = a_1 + b_1 a_s^{-1} \quad (24)$$

where

$$a_1 = \frac{8760 n_t C_e A D}{u(UF)(DF)} \left[b' + d'(UF)^2 (LLF) \right]$$

$$b_1 = \frac{8760 n_t C_e A}{u} \left[a' + c' (UF)^2 (LLF) \right]$$

Feeder Loss Cost

Considering the feeder service area, a_f to be rectangular with dimensions $L \times 2b$, having uniformly distributed loads along the feeder, the total power loss in all the secondary mains in the system can be obtained as in (25), which is derived in Appendix B.

$$P_{ms} = k_2 L a_f \quad (25)$$

where

$$k_2 = rA \left[\frac{D}{(kV)(PF)(DF)} \right]^2 \left[\frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2} \right] \quad (26)$$

Let the length of each lateral be b km and let each lateral carry a load of p kW, where

$$b = \frac{a_f}{2L} \quad (27)$$

$$p = \frac{a_f D}{2L (DF)} \quad (28)$$

Total power loss in all the secondary laterals can be obtained as

$$P_{ss} = k_2' \left(\frac{1^2}{8L^3} \right) a_f^2 \quad (29)$$

In (29) k_2' will be same as k_2 in (26) except that n in this case is different. In the case of continuously uniformly distributed loads $k_2' = k_2$.

The total power loss in the system is, thus, the sum of P_{ms} and P_{ss} and the corresponding energy loss is

$$E_{s1} = 8.76 (LLF) \left(k_2 L a_f + \frac{k_2' 1^2}{8 L^3} a_f^2 \right) \quad (30)$$

The cost of this energy loss is a recurring expenditure annually. This annual expenditure is capitalized by dividing the same with a capitalization factor, u , which includes depreciation, interest and operation and maintenance charges and the capitalized cost is given as

$$C_{s1} = \frac{E_{s1} C_e}{u} \quad (31)$$

The primary distribution feeder losses and their capitalized cost are obtained in a similar way. The only difference will be that the primary feeder length is also a function of the radius of the secondary distribution substation feed area. The length of the lateral, b is given in terms of primary main length and the secondary radius, R_s as

$$b = \frac{a_f}{2(L + R_s)} - R_s \quad (32)$$

Incorporating this change, the capitalized cost of energy losses in the primary distribution feeders is obtained as

$$C_{s1}^p = \frac{8.76 (LLF) C_e}{u} \left[\left(L - \frac{R_s^3}{L^2} \right) k_2 a_f + \frac{k_2' R_s^2}{2L^2(L + R_s)} a_f^2 \right] \quad (33)$$

OBJECTIVE FUNCTIONS

An objective function, F_s is defined for substation feed area, assuming constant voltage drop in the

feeders. Another function, F_v is defined for the feeder area keeping the substation area constant. The optimized feeder area in turn gives the optimal load carrying level of the feeder.

The function, F_s constitutes the capital investment for creating new facilities and the capitalized cost of energy losses in the system, as given in (34)

$$F_s = (\text{Capital investment required for feeders}) + (\text{Capital investment required for substations}) + (\text{Capital investment required for infeed circuits to the substations}) + (\text{Capitalized cost of transformation energy losses}) + (\text{Capitalized cost of energy losses in the feeders}) \quad (34)$$

If the voltage drop is kept constant in the feeders, the total energy losses in the system will remain constant irrespective of the substation feed area (see Appendix B), and therefore, the term corresponding to the feeder loss cost may be dropped from equation (34). The investment for infeed circuits to the substation is a function of the radius of the substation feed area and the number of source stations, N , available to feed the substations. Each substation, except those at the source stations, would require a length equal to twice the radius of the substation feed area. F_s is expressed as a function of n_f , n_s , L , N and kVA as

$$F_s = c_f L n_f n_s + (e + g \cdot kVA + f n_f) n_s + \frac{2L}{z} (n_s - N) c_{fe} + (a_1 + \frac{b_1}{A} n_s) \quad (35)$$

The variables n_f and L can be shown as functions of a_s as in (36) and (37) respectively.

$$n_f = \left[\frac{1}{\pi^{3/2} k_1} \right] a_s^{3/2} \quad (36)$$

$$L = \left(\frac{z}{\pi^{1/2}} \right) a_s^{1/2} \quad (37)$$

F_s can be represented as a function of a_s by substituting for n_f , n_s , L and kVA from (36), (23), (37) and (22) respectively as

$$F_s = \frac{c_f G_1 G_3}{A} a_s + \left(\frac{f G_1}{A^{1/2}} - G_5 \right) a_s^{1/2} + (a_1 + g G_2) + G_4 a_s^{-1/2} + (eA + b_1) a_s^{-1} \quad (38)$$

where

$$G_1 = \frac{A^{3/2}}{k_1^3 \pi^{3/2}} ; G_3 = z k_1 G_1^{1/3}$$

$$G_2 = \frac{A D}{(UF)(DF_s)(PF)} ; G_4 = \left(\frac{2A}{\pi^{1/2}} \right) c_{fe} ; G_5 = \left(\frac{2N}{\pi^{1/2}} \right) c_{fe}$$

The optimal feed area, a_s^* is obtained from (38), solving (39)

$$a_s^{*2} + \frac{A}{2c_f G_1 G_3} \left(\frac{f G_1}{A^{1/2}} - G_5 \right) a_s^{*3/2} - \frac{A G_4}{2c_f G_1 G_3} a_s^{*1/2} = \frac{(eA + b_1) A}{c_f G_1 G_3} \quad (39)$$

The function, F_v is used to optimize a_f at constant a_s . If a_s is kept constant, all the terms in

(34), except two terms corresponding to feeder cost and feeder loss cost would remain constant in the case of secondary system; because the bay cost can be neglected (i.e. $f = 0$). Thus, F_{vs} for one secondary distribution substation area is obtained as

$$F_{vs} = c_f L n_f + C_{s1}/n_s \quad (40)$$

The feeder main length, L , will remain constant since a_s is constant. Defining F_{vs} as a function of a_f using (30) and (31), we get,

$$F_{vs} = c_f L a_s a_f^{-1} + \frac{8.76 k(LLF)C_e}{u n_s} \left[L a_f + \frac{1^2}{8L^3} a_f^2 \right] \quad (41)$$

The optimal feed area of the secondary feeder, a_f^* can be obtained by solving (42), which is derived from (41)

$$a_f^{*3} + \frac{4L^4}{1^2} a_f^{*2} = \frac{c_f A u L^4}{2.19k(LLF)C_e 1^2} \quad (42)$$

where $k = k_2 = k_2'$ for continuously uniformly loaded system.

In the case of primary distribution systems, the cost cannot be neglected (f will be nonzero) and, therefore, substation cost will remain constant with varying a_f^* . Thus, F_{vp} will be a function of the feeder bay cost feeder loss cost and feeder cost as

$$F_{vp} = c_f L n_f + f n_f + C_{s1}^p/n_s \quad (43)$$

Representing F_{vp} as a function of a_f using (33), we get,

$$F_{vp} = (c_f L + f) a_s a_f^{-1} + \frac{8.76 k(LLF)C_e}{u n_s} \left[\left(L - \frac{R_s^3}{L^2} \right) a_f + \frac{R_s^2}{2L^2(L+R_s)} a_f^2 \right] \quad (44)$$

The optimal feed area of the primary feeder, a_f^* can be obtained by solving (45) derived from (44)

$$a_f^{*3} + \frac{(L^3 - R_s^3)(L+R_s)}{R_s^2} a_f^{*2} = \frac{(c_f L + f)L^2(L+R_s)A u}{8.76 k(LLF)C_e R_s^2} \quad (45)$$

ALGORITHM DESCRIPTION

From the above discussion it can be seen that the primary aim of this work is to optimize the substation size (or the substation feed area), feeder loading limit (or the feeder area) and conductor sizes for feeders. Equations (39) with $f = 0$ and (42) for secondary system and (39) and (45) for primary system will give the necessary equations for this purpose. In equation (39) a_s is optimized with constant voltage drop, v , whereas in (42) or (45) a_f is optimized with constant a_s . Since a_s^* is a function of v , a_f^* , obtained from (42) or (45) will be optimal only when the voltage drop corresponding to the optimized a_f^* is the same as that assumed in (39) while solving for a_s^* . It is, therefore, necessary to obtain a_s^* and a_f^* for the same v so as to ensure optimality. This is achieved through a two level optimization procedure solving for a_s^* and a_f^* iteratively until a_s^* and a_f^* are obtained for the same percentage voltage regulation.

Since, the conductor size is assumed constant

throughout in the models discussed, minimum cost solutions for different standard conductor sizes can be generated with the help of the defined objective functions and from a comparison of these minimum costs, the most economical conductor size can then be selected.

Further, in formulating general guidelines on planning policies for future systems under varying load densities, the optimum distribution system parameters can be generated for several load densities with the help of the objective functions discussed above.

The steps of the proposed algorithm are described below :

Steps

1. Compute the coefficients in (39) with the help of system data selecting a value for v , arbitrarily.
2. obtain a_s^* from (39)
3. Using a_s^* , computed in step 2, and the other system data, calculate the coefficients in (42) or (45) depending upon whether the system optimized is secondary or primary.
4. obtain a_f^* from (42) or (45)
5. Estimate the percentage voltage drop corresponding to a_f^* , obtained in step 4.
6. With the new value of v , go to step 2 and repeat the steps 2 to 5 until the change in v is less than a preselected tolerance (0.001).
7. Using the optimal values obtained for substation feed area and feeder area, compute the corresponding optimal substation capacity, number of feeders per substation and the load carrying limit of the feeders for the system under study with a given load density.
8. Obtain minimum cost solutions for different conductor sizes by repeating the steps 1 to 7 and select the most economical conductor size.
9. Repeat the operations of steps 1 to 8 for different expected load densities in the system at a future date for determining the corresponding optimal system parameters.

RESULTS AND CONCLUSION

The proposed method has been tried on a typical distribution system in India, assuming uniformly distributed loads with an LDF of 1.8 at 0.8 power factor. The zig-zag factor, z for the area was taken as 1.5 as observed in the system. The information of the substation cost and the transformation losses has already been presented earlier in the paper. The cost data for the feeders with different conductor sizes assumed in this study are given in Table 1.

The optimal parameters are obtained using the suggested algorithm for both secondary and primary distribution systems for load densities varying from 5 kW/sq.km to 100 kW/sq.km and are presented in Tables 2 and 3. The results obtained suggest that this study area should have small size transformers with short feeders as against the present practice in the area. It may be noted here that this aspect would have been lost sight off in the normal simulation process if the small size transformers were not considered.

Table 1
Feeder Costs

Sl. No.	Conductor Size	Cost per km in Rs. ($\times 10^5$)
1	Squirrel (13 mm ² ACSR)	0.1044
2	Gopher (16 mm ² ACSR)	0.1188
3	Weasel (20 mm ² ACSR)	0.1380
4	Ferret (25 mm ² ACSR)	0.1620
5	Rabbit (30 mm ² ACSR)	0.1860

Table 2

Optimal Parameters for Secondary System

Sl.No.	Load density (kW/sq.km)	Feeder Area (sq.km)	Substation capacity (kVA)	Conductor size
1	5	2.107	15	Squirrel
2	10	1.542	20	Squirrel
3	15	1.283	25	Squirrel
4	20	1.067	25	Gopher
5	30	0.858	30	Gopher
6	40	0.737	35	Ferret
7	50	0.649	40	Rabbit
8	100	0.349	45	Rabbit

Table 3

Optimal Parameters to Primary Systems

Sl.No.	Load density (kW/sq.km)	Feeder area (sq.km)	Substation capacity (kVA)	Conductor size
1	5	48.077	535	Rabbit
2	10	35.210	780	Rabbit
3	15	29.070	970	Rabbit
4	20	25.773	1145	Rabbit
5	30	21.552	1435	Rabbit
6	40	19.231	1710	Rabbit
7	50	17.606	1955	Rabbit
8	100	13.298	2955	Rabbit

The results for both secondary and primary distribution systems for the densities listed in the tables have been obtained in about 18 seconds of cpu time

on IBM 360/44 computer system. The time required for obtaining the optimal results for either secondary or primary system for one load density is about 3 seconds. The time taken by the simulation process for producing similar results for both primary and secondary systems for one load density is about 10 minutes and it is nearly an hour for all the densities listed in Table 2. The computational time for the proposed method is extremely less compared to the simulation technique.

Using the suggested models, the system planners can carryout a sensitivity analysis, with respect to the various component costs such as substation and feeder costs. This analysis would enable the choice of cheaper design of lines or substations. The proposed method is very efficient, fast and accurate in obtaining the optimal distribution system parameters. Further, it is simple and easy to program and hence, in the authors' view, is highly promising.

NOTATIONS

A	area of the study system in sq.km
D	load density in kW/sq.km
a_s	feed area of the substation in sq.km
a_f	feeder area in sq.km
n_s	number of substations in the area A
n_f	number of feeders per substation
R	radius of the circular feed area of the primary distribution substation in km
R_s	radius of the circular feed area of the secondary distribution substation in km
P	total load in the feeder in kVA
p	lateral load in kW
R_i	resistance of the i th section of the feeder in Ohms
R_T	total resistance of the feeder in Ohms
I_T	total current in the feeder in Amps
kVA_i	power flow in kVA in the i th section of the feeder
km_i	distance of the i th section of the feeder in km
r	resistance of the feeder in Ohms per km
x	reactance of the feeder in Ohms per km
θ	power factor angle
v	percentage voltage regulation of the feeder
kV	circuit voltage in kilo Volts
PF	power factor of the load
LF	load factor
LLF	loss load factor which is a function of LF of the form $LLF = A(LF)^2 + B(LF)$, where $A + B = 1$
UF	utilization factor of the transformers
DF	diversity factor representing the diversity of customer loads at the feeders
DF_s	diversity factor of customer loads at the substation
n_t	number of transformers at each substation
L	length of the feeder main
c_f	feeder cost in Rs. per km
C_e	cost of energy in Rs. per unit
C_{fe}	infeed circuit cost in Rs. per km
u	capitalization factor

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APPENDIX A

Load Distribution Factor (LDF), for uniformly discretely loaded radial feeders like the one in Fig.(a) can be obtained from its definition given in (A.1).

$$LDF = \frac{PL}{M} = \frac{PL}{Hv} \quad (A.1)$$

where H is the unit voltage regulation constant for a given conductor size, power factor and circuit voltage; v is the percentage voltage drop in the feeder. P is the total load in the feeder; L is the feeder length and M is the moment of the loads.

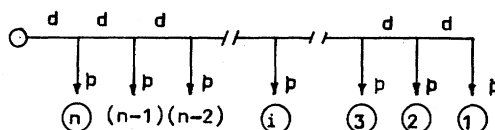


Fig.(a) Uniformly loaded radial feeder

In the feeder in Fig.(a) the total load,

$$P = \frac{np}{(PF)} \quad (A.2)$$

The length of the feeder,

$$L = nd \quad (A.3)$$

The moment of the loads,

$$\begin{aligned} M &= \left(\frac{pd}{(PF)} \right) (1 + 2 + 3 + \dots + n) \\ &= \left(\frac{pd}{(PF)} \right) \frac{n(n+1)}{2} \end{aligned} \quad (A.4)$$

Substituting for P, L and M from (A.2), (A.3) and (A.4) in (A.1), we get,

$$LDF = \frac{2n}{n+1} \quad (A.5)$$

Similarly, for a case of non-uniformly loaded feeder as in Fig. (b)

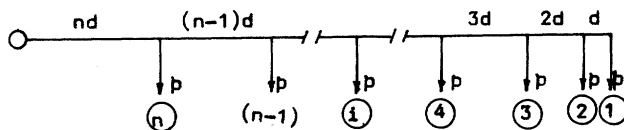


Fig.(b) Non-uniformly loaded feeder - Case (i)

$$P = \frac{np}{(PF)} \quad (A.6)$$

$$L = \frac{n(n+1)}{2} d \quad (A.7)$$

$$\begin{aligned} M &= \left(\frac{pd}{(PF)} \right) (1^2 + 2^2 + 3^2 + \dots + n^2) \\ &= \left(\frac{pd}{(PF)} \right) \frac{n}{6} (n+1)(2n+1) \end{aligned} \quad (A.8)$$

From (A.6), (A.7), (A.8) and (A.1) LDF can be obtained as

$$LDF = \frac{3n}{2n+1} \quad (A.9)$$

Following the same procedure, for the type of non-uniform loading shown in Fig. (c) (Case (ii)),

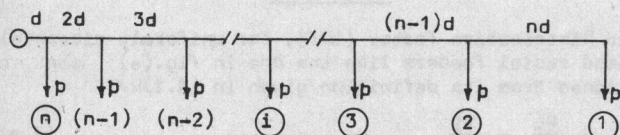


Fig.(c) Non-uniformly loaded feeder - Case (ii)

$$P = \frac{np}{(PF)} \quad (A.10)$$

$$L = \frac{n(n+1)}{2} d \quad (A.11)$$

$$M = \frac{pd}{(PF)} (n.1 + (n-1).2 + \dots + i.(n-i+1) + \dots + 2.(n-1) + 1.n) \quad (A.12)$$

$$= \frac{pd}{(PF)} \sum_{i=1}^n [i.(n-i+1)]$$

and

$$LDF = \frac{n^2(n+1)}{2 \sum_{i=1}^n [i(n-i+1)]} \quad (A.13)$$

APPENDIX B

Computation of Feeder Losses :

Consider a radial feeder of the type shown in Fig. (a) (Appendix A) with uniformly distributed loads. If the voltage magnitude at all the lateral take-off points is assumed to be the same, the lateral load currents are all equal. Letting the sectional resistance of the feeder be R_1 and the lateral current I , the total power loss in the feeder can be obtained as

$$P_L = 3 I^2 R_1 + 3 (2I)^2 R_1 + 3 (3I)^2 R_1 + \dots + 3 \left[(n-1) I \right]^2 R_1 + 3 (nI)^2 R_1 \quad (B.1)$$

Simplifying (B.1) and introducing I_T for total current (nI) in the feeder and R_T for the total feeder resistance $(n R_1)$, (B.1) reduces to (B.2)

$$P_L = 3 I_T^2 R_T \left(\frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2} \right) \quad (B.2)$$

Replacing I_T in terms of feeder peak load P in kVA, we get

$$P_L = \left[\frac{P}{(kv)} \right]^2 R_T \left(\frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2} \right) \quad (B.3)$$

Substituting for P in terms of feeder area and load density into (B.3), P_L is obtained as a function of a_f as

$$P_L = \left[\frac{D}{(kv)(DF)} \right]^2 R_T \left[\frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2} \right] a_f^2 \quad (B.4)$$

If the load is continuously uniformly distributed (i.e. for $n = \infty$), P_L is given by

$$P_L = \frac{1}{3} R_T \left[\frac{D}{(kv)(DF)} \right]^2 a_f^2 \quad (B.5)$$

It can be shown that in a distribution system, the total losses remain constant irrespective of the number of substations and feeders, if the feeder voltage regulation is kept constant.

The total power loss in the system is

$$P_{TL} = P_L n_f n_s \quad (B.6)$$

Substituting for P_L from (B.5) and replacing n_f and n_s in terms of the feeder area, a_f , we get,

$$P_{TL} = k' L D a_f \quad (B.7)$$

$$\text{where } k' = \frac{1}{3} r AD \left[\frac{1}{(kv)(DF)} \right]^2 \quad (B.8)$$

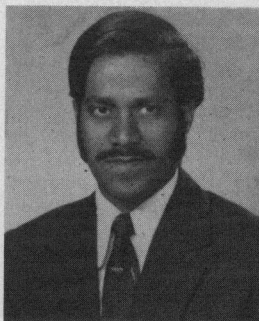
The feeder load,

$$P = \frac{D a_f}{(PF)} \quad (B.9)$$

From (B.7), (B.9) and (A.1), we get,

$$P_{TL} = k' (LDF) (PF) H v = k'' v \quad (B.10)$$

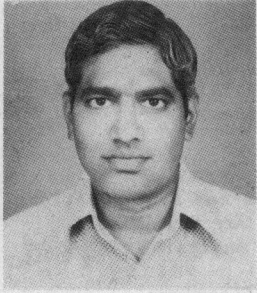
In (B.10) H is constant for given conductor size, power factor and circuit voltage; LDF is constant for a particular type of distribution pattern of loads; and k' is constant for a particular circuit voltage, power factor, load diversity, load density, study area A , and the resistance in Ohms per km of the line. Thus, for constant k'' , the total power loss P_{TL} will remain constant, if the voltage regulation, v is kept constant.



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